

National Higher School of Mathematics

Preparatory Cycle
First Year

Academic Year: 2024/2025
Course: Algebra 1

Worksheet 4

Algebraic Structures

The starred exercises are to be done in tutorial sessions

Groups

Exercise 1. ★

On $\mathbb{R}$, we define the binary operation $*$ by: $\forall x, y \in \mathbb{R}, x * y = x + y + \sin(xy)$.

1. Is this binary operation commutative?
2. Does it have a neutral element?
3. Let $f(x) = x + 3 + \sin(3x)$. Compute $f(-4)$, $f(-3.8)$, and $f(-3.1)$.
4. What can we deduce from this?
5. Deduce that $*$ is not associative.

Exercise 2.

1. On the set H of all binary relations on a set E , we define the binary operation $*$ by: $R * S = T$, where T is defined as follows:

$$\forall x, y \in E, xTy \iff \exists z \in E, \text{ such that } (xRz) \text{ and } (zSy).$$

Show that $*$ is associative.

2. We define on $\mathbb{Z}$ the binary operation $*$ by:

$$x * y = xy + \lfloor \phi x \rfloor \lfloor \phi y \rfloor,$$

where ϕ denotes the golden ratio. Is $*$ associative?

Exercise 3. ★

1. Show that $(G, *)$ is a group in the following cases:
 - (a) $G = \mathbb{R} \times \mathbb{R} \setminus \{(0, 0)\}$, $(a, b) * (c, d) = (ac - bd, bc + ad)$.
 - (b) $G =]-1, 1[$, $x * y = \frac{x + y}{1 + xy}$.
2. Let E be a nonempty set. Show that $(\mathcal{P}(E), \Delta)$ is an abelian group.

Exercise 4. ★

1. We consider the set $S = \{f_1, f_2, f_3, f_4, f_5, f_6\}$ of maps from $\mathbb{R} \setminus \{0, 1\}$ into $\mathbb{R} \setminus \{0, 1\}$, where $f_1(x) = x$, $f_2(x) = 1 - x$, $f_3(x) = \frac{1}{x}$, $f_4(x) = \frac{1}{1-x}$, $f_5(x) = \frac{x-1}{x}$ and $f_6(x) = \frac{x}{x-1}$.
 - (a) Write the multiplication table of S and show that $(S, \circ)$ is a group.
 - (b) What are the subgroups of S ?
2. Write the multiplication table of a group of order 2 and a group of order 3.

Exercise 5 (Minimization of group axioms).

Let G be a nonempty set equipped with a binary operation $*$, which is associative, has a right neutral element $e \in G$, and each element of G admits a right symmetric.

Show that $(G, *)$ is a group.

Exercise 6.

Let $(G, \cdot)$ be a group and e be its neutral element.

1. Show that for any $a \in G$ and $n, m \in \mathbb{Z}$, we have $a^n a^m = a^{n+m}$.
2. Show that if every element of G is its own inverse, then G is abelian.
3. Assume that G is of even order. Prove that G has an element $a \neq e$ such that $a^2 = e$.
4. Assume that G is finite. Show that there exists a positive integer m such that $a^m = e$ for all $a \in G$.
5. Show that if for all $a, b \in G$, $(ab)^2 = a^2 b^2$; then G is abelian.

Exercise 7.

Let $(G, \cdot)$ be a group.

1. Show that for all $a, b \in G$, the elements ab and ba have the same order.
2. Suppose $(G, \cdot)$ is finite and abelian. Let $x, y \in G$ with respective orders n and m . Show that if n and m are coprime, then xy is of order mn .
3. Assume that an element x of G has order rs . Find the order of x^r .
4. Assuming that x has n as order, what is the order of x^r ?

Exercise 8. ★

We define on $\mathbb{R}^2$ the binary operation $\oplus$ as follows:

$$\forall (x, y), (x', y') \in \mathbb{R}^2, (x, y) \oplus (x', y') = (x + x', y + y' + 2xx').$$

1. Show that $(\mathbb{R}^2, \oplus)$ is an abelian group.
2. Show that $H = \{(x, x^2) \mid x \in \mathbb{R}\}$ is a subgroup of $(\mathbb{R}^2, \oplus)$.
3. Show that the map $\Phi : (\mathbb{R}, +) \longrightarrow (H, \oplus)$ defined by $\Phi(x) = (x, x^2)$ is an isomorphism of groups.

Exercise 9 (Remarkable subgroups). ★

Let $(G, \cdot)$ be a group, H and K be two subgroups of G , and $a \in G$. Show that the following sets are subgroups of G .

1. $H \cap K$.
2. $Z(G) = \{x \in G \mid \forall y \in G, xy = yx\}$ ($Z(G)$ is called the center of G).
3. $CG(a) = \{x \in G \mid xa = ax\}$ ($CG(a)$ is called the centralizer of a in G).
4. $aHa^{-1} = \{aha^{-1} \mid h \in H\}$.
5. $N(H) = \{g \in G \mid gHg^{-1} = H\}$ ($N(H)$ is called the normalizer of H in G).

Exercise 10.

1. Prove that the group $(\mathbb{R}, +)$ is not cyclic.
2. Show that any subgroup of a cyclic group is cyclic and give an example of a non-cyclic group whose proper subgroups are all cyclic.
3. Let p and q two primes. Show that if G is a finite group of order pq , then every proper subgroup of G is cyclic.

Exercise 11.

Let H and K be two subgroups of a group G .

1. Show that $H \cup K$ is a subgroup of G if and only if $H \subset K$ or $K \subset H$.
2. We define HK by $HK = \{hk \mid h \in H, k \in K\}$. Show that HK is a subgroup of G if and only if $HK = KH$.

Exercise 12 (Classification of subgroups of $(\mathbb{R}, +)$).

Let G be a subgroup of $(\mathbb{R}, +)$ such that $G \neq \{0\}$. In this exercise, we will show that G is either dense in $\mathbb{R}$ or there is $\alpha \in \mathbb{R}$ such that $G = \alpha\mathbb{Z}$.

1. Show that the set $G \cap \mathbb{R}_+^*$ is nonempty.
2. We put $\alpha = \inf(G \cap \mathbb{R}_+^*)$. We suppose that $\alpha > 0$.
 - (a) Show that $\alpha \in G$ and deduce that $\alpha\mathbb{Z} \subset G$.
 - (b) Show that for all $g \in G$, we have $g = \alpha E\left(\frac{g}{\alpha}\right)$.
 - (c) Deduce that $G = \alpha\mathbb{Z}$.
3. Suppose that $\alpha = \inf(G \cap \mathbb{R}_+^*) = 0$.
 - (a) Show that for all $x, y \in \mathbb{R}$, we have: $x < y \implies \exists g \in G, x < \left(E\left(\frac{x}{g}\right) + 1\right)g < y$.
 - (b) Deduce that G is dense in $\mathbb{R}$.

Exercise 13 (On the subgroups of S^1). ★

Let n be a positive integer. Let $\mathbb{U} = \{z \in \mathbb{C} \mid |z| = 1\}$ be the unit circle, $\mathbb{U}_n = \{z \in \mathbb{C} \mid z^n = 1\}$ the set of n th roots of unity, and $\mathbb{U}_\infty = \bigcup_{n \in \mathbb{N}^*} \mathbb{U}_n$ the set of all the n th roots of unity.

1. Show that $\mathbb{U}$ and $\mathbb{U}_n$ are subgroups of $(\mathbb{C}^*, \times)$. Is this the case for $\mathbb{U}_\infty$?
2. Show that $\mathbb{U}_n \subset \mathbb{U}_m$ if and only if $n \mid m$.
3. Show that $\mathbb{U}_n$ is cyclic.

II. Let $\alpha \in \mathbb{R}$ and consider the map

$$\begin{aligned} f_n: \mathbb{Z} &\longrightarrow \mathbb{C}^* \\ p &\longmapsto f_n(p) = e^{2i\pi p n \alpha}. \end{aligned}$$

1. Show that f_n is a group morphism from $(\mathbb{Z}, +)$ to $(\mathbb{C}^*, \times)$.
2. Show that $\text{Im } f_n \subset \mathbb{U}$.
3. Show that f_n is injective if and only if $\alpha \notin \mathbb{Q}$.

III. Consider the map

$$\begin{aligned} \phi: \mathbb{R} &\longrightarrow \mathbb{U} \\ t &\longmapsto \phi(t) = e^{it}. \end{aligned}$$

1. Show that ϕ is a group morphism from $(\mathbb{R}, +)$ to $(\mathbb{U}, \times)$. Is ϕ injective? Surjective?
2. We define on $\mathbb{R}$ the binary relation $\mathcal{R}$ by: $\forall x, y \in \mathbb{R}, x \mathcal{R} y$ if and only if $\exists k \in \mathbb{Z}, x - y = 2k\pi$. Show that $\mathcal{R}$ is an equivalence relation.
3. On $\mathbb{R}/\mathcal{R}$ we define the binary operation $\oplus$ by: $\bar{x} \oplus \bar{y} = \overline{x+y}$ for $\bar{x}, \bar{y} \in \mathbb{R}/\mathcal{R}$. Show that $(\mathbb{R}/\mathcal{R}, \oplus)$ is a group isomorphic to $(\mathbb{U}, \times)$.

Exercise 14. ★

On $\mathbb{R}$, we define the binary operation $*$ as follows :

$$\forall x, y \in \mathbb{R}, x * y = \sqrt[3]{x^3 + y^3}.$$

Show that $(\mathbb{R}, *)$ is an abelian group isomorphic to $(\mathbb{R}, +)$.

Exercise 15. ★

- I
 1. Determine all the group homomorphisms from $(\mathbb{Z}, +)$ into $(\mathbb{Z}, +)$. Which are injective? surjective?
 2. Same question for the group homomorphisms from $(\mathbb{Q}, +)$ into $(\mathbb{Z}, +)$.
- II Show that the following statements are true:
 1. There is no group isomorphism between $(\mathbb{R}, +)$ and $(\mathbb{R}^*, \times)$.
 2. There is no group isomorphism between $(\mathbb{Q}, +)$ and $(\mathbb{Q}_+^*, \times)$.

Exercise 16.

Let $(G, *)$ be a group and $\text{Aut}(G)$ be the set of automorphisms of G .

1. Show that $(\text{Aut}(G), \circ)$ is a group.
2. Let H be a subgroup of $\text{Aut}(G)$, and let ϕ be the following map:

$$\begin{aligned} \phi: G &\longrightarrow \mathcal{P}(G) \\ x &\longmapsto \{f(x) \mid f \in H\}. \end{aligned}$$

$\phi(x)$ is called the orbit of x under H . Show that $\phi(G)$ is a partition of G .

Rings

Exercise 17. ★

Complete the operation tables for the ring $R = \{a, b, c, d\}$:

+	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	a	b
d	d	c	b	a

*	a	b	c	d
a	a	a	a	a
b	a	b		
c	a			a
d	a	b	c	

Is R a commutative ring? What is its characteristic? (Hint: $c * b = (b + d) * b$, $c * c = c * (b + d)$, ..., etc).

Exercise 18.

Let $(R, +, \times)$ be a commutative ring, and N the set of nilpotent elements of R . Let $B = \{1 + x \mid x \in N\}$. Show that $(B, \times)$ is a group.

Exercise 19. ★

Let $\mathbb{Z}/n\mathbb{Z}$ be the set of integers modulo n .

- I.
 1. Verify that $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring.
 2. What is the characteristic of the ring $\mathbb{Z}/5\mathbb{Z}$? and the ring $\mathbb{Z}/6\mathbb{Z}$?
 3. Show that $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.

II. Let m and n be two integers. Put $m\mathbb{Z} + n\mathbb{Z} = \{an + bm \mid a, b \in \mathbb{Z}\}$. Show that :

1. $m\mathbb{Z} + n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.
2. m and $n + 7m$ generate the subgroup $m\mathbb{Z} + n\mathbb{Z}$.

III. Let m and n be two positive integers.

1. Show that $m\mathbb{Z} \cap n\mathbb{Z} = \text{lcm}(n, m)\mathbb{Z}$.
2. Is $2\mathbb{Z} \cup 3\mathbb{Z}$ a subring of $\mathbb{Z}$?
3. Find a necessary and sufficient condition on m and n for which $m\mathbb{Z} \cup n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.
4. Determine the smallest subgroup of $\mathbb{Z}$ containing $m\mathbb{Z}$ and $n\mathbb{Z}$.

Exercise 20. ★

We define on $\mathbb{C}$ the binary operations $\oplus$ and $\otimes$ as follows : for any $z = a + ib$, $z' = a' + ib' \in \mathbb{C}$,

$$z \oplus z' = (a + a') + i(b + b') \text{ and } z \otimes z' = aa' + i(ab' + ba').$$

1. Show that $(\mathbb{C}, \oplus, \otimes)$ is a commutative ring.
2. Determine the set U of invertible elements in $(\mathbb{C}, \oplus, \otimes)$.
3. Determine the set D of zero divisors in $(\mathbb{C}, \oplus, \otimes)$.
4. Let $z = a + ib \in \mathbb{C}$. For all $n \in \mathbb{N}$, let : $z^{(n)} = \underbrace{z \otimes z \otimes z \cdots \otimes z}_{n \text{ times}}$.

- (a) Write $z^{(n)}$ as a function of n, a , and b .
- (b) Solve in $\mathbb{C}$ the equation $z^{(4)} \oplus z^{(2)} = 6 + i$.

Exercise 21.

On $\mathbb{R}^2$, consider the addition

$$(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$$

and the multiplication

$$(x_1, x_2) \times (y_1, y_2) = (x_1 y_1 - x_2 y_2, x_1 y_2 + x_2 y_1).$$

1. Verify that $(\mathbb{R}^2, +, \times)$ is ring.
2. Find the zero divisors of $(\mathbb{R}^2, +, \times)$.

Exercise 22.

Let $R = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$, and for all $x = a + b\sqrt{2}$ of R , let $N(x) = a^2 - 2b^2$.

1. Show that R is a subring of $\mathbb{R}$.
2. Show that for all x, y of R , $N(xy) = N(x)N(y)$.
3. Deduce that x is invertible in R if and only if $N(x) = \pm 1$.
4. Show that the elements $\pm(1 + \sqrt{2})^n$, $n \in \mathbb{Z}$, are invertibles in R .
5. Conversely, we want to show that any invertible element x of R is of such form.
 - (a) Show that we can assume that $x = a + b\sqrt{2}$, with $a \in \mathbb{N}$ and $b \in \mathbb{N}_0$.
 - (b) Then show that x is of the form $(1 + \sqrt{2})^n$ with $n \in \mathbb{N}_0$. Conclude.

Hint: if $b \geq 1$, consider $x_1 = \frac{x}{1 + \sqrt{2}}$.

Exercise 23. ★

Let R be the set defined as follows:

$$R = \left\{ \frac{a}{10^n} \mid a \in \mathbb{Z}, n \in \mathbb{N} \right\}.$$

1. Show that R is a subring of $\mathbb{Q}$ for the usual binary operations.
2. Show that 3 has no inverse in R . Conclude.
3. Determine the set of invertible elements in R .

Exercise 24.

1. Let R be a ring and $C = \{x \in R \mid \forall y \in R, xy = yx\}$ (C is called the center of R). Show that C is a subring of R .
2. Determine the smallest subring of $\mathbb{Q}$ that contains $\frac{1}{2}$.
3. (a) Find all ring homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}/30\mathbb{Z}$.
(b) In each case describe the kernel and the image.

Exercise 25 (Particular ideals).

Let R be a ring with unity.

1. The sum of two ideals I and J is the set defined by $I + J = \{i + j \mid i \in I \text{ and } j \in J\}$. Show that $I + J$ is also an ideal.
2. Suppose that R is commutative, and let I and J be two ideals of R . We define the following set:

$$[I : J] = \{r \in R \mid \forall s \in J, rs \in I\}.$$

Show that $[I : J]$ is an ideal ($[I : J]$ is called the quotient ideal of I by J).

3. Suppose that R is commutative.
(a) Let J be an ideal of R . We define the radical of J as follows :

$$\sqrt{J} = \{a \in R \mid r^n \in J \text{ for some } n \in \mathbb{N}\}.$$

Show that $\sqrt{J}$ is an ideal of R .

- (b) Deduce that the set of nilpotent elements

$$I = \{x \in R \mid x^n = 0 \text{ for some integer } n\}$$

is an ideal of R (I is called the nilradical of R).

- (c) Let A be any subset of R . The annihilator of A is defined by:

$$\text{Ann}(A) = \{r \in R \mid ra = 0 \text{ for all } a \in A\}.$$

Show that $\text{Ann}(A)$ is an ideal of R .

Exercise 26 (Boolean ring). ★

Let R be a ring, in which, for any element x , we have $x^2 = x$.

1. Give examples of such rings.
2. Show that for any a , we have $2a = 0$.
3. Deduce that R is commutative.
4. Show that R cannot be reduced to three elements.
5. Assume that R is finite and of cardinal greater than 2. Show that R has divisors of zero (Hint: Consider the element $xy(x + y)$).
6. Show that if $\text{card}(R) = 4$, then R is unique up to an isomorphism.

7. Show that if R is finite, then its cardinal is a power of 2.

Exercise 27 (Euclidean rings).

Let R be a ring. We say that R is an euclidean ring if R is an integral domain, and there is a map $v : R \setminus \{0\} \rightarrow \mathbb{N}$, such that:

$$\forall a, b \in (R \setminus \{0\}), \exists q, r \in R : a = qb + r \text{ and } (r = 0 \text{ or } v(r) < v(b)).$$

1. Show that $\mathbb{Z}$ and $\mathbb{R}[x]$ are euclidean rings.
2. Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$. By considering the map $v : z \mapsto |z|^2$, show that $\mathbb{Z}[i]$ is an euclidean ring.

$\mathbb{Z}$ and its arithmetic

Exercise 28 (Fermat numbers).

For any nonnegative integer n , let $F_n = 2^{2^n} + 1$.

1. Show that F_n divides $F_{n+k} - 2$ for any nonnegative integer n and any positive integer k .
2. Let n, m , and d be three nonnegative integers such that $n \neq m$. Show that if d divides both F_n and F_m , then $d = 1$.

Exercise 29. ★

1. Show that $\forall (a, b) \in \mathbb{Z}^2$, if 7 divides $a^2 + b^2$, then 7 divides both a and b .
2. Find all positive integers x such that $2^{2^x+1} + 2$ is divisible by 17.
3. Find all integers x such that $x^{86} \equiv 6 \pmod{29}$.
4. Solve in $\mathbb{Z}$ the systems :

$$\begin{cases} x \equiv 2 \pmod{6} \\ x \equiv 5 \pmod{9} \end{cases} \quad \text{and} \quad \begin{cases} x \equiv 2 \pmod{5} \\ x \equiv 3 \pmod{7} \\ x \equiv 10 \pmod{11} \end{cases}.$$

Exercise 30.

1. Find all positive integer solutions (x, y, z) of the following equations

$$x^3 = 3^y 7^z + 8 \quad \text{and} \quad 3^x + 4^y = 5^z.$$

2. Find all integers $a, b, c, d \geq 0$ such that

$$4^a + 5^b + 6^c = 7^d.$$

3. Prove that there is no $n \in \mathbb{N}$ such that $2n - 1, 5n - 1, 13n - 1$ are all perfect squares.

Exercise 31.

We define on $\mathbb{R}$ the binary operations $\oplus$ and $\otimes$ as follows: $\forall x, y \in \mathbb{R}, x \oplus y = x + y - 1$ and $x \otimes y = x + y - xy$. Show that $(\mathbb{R}, \oplus, \otimes)$ is a field.

Exercise 32. ★

1. Prove that any field homomorphism is a monomorphism.
2. Prove that any field is an integral domain.
3. If F is a field, prove that its only ideals are (0) and F itself.
4. Show that a finite integral domain is a field.
5. Find all the subfields of $\mathbb{Q}$.
6. Can we find a field for which the group $(F, +)$ is isomorphic to the group $(F^*, \times)$?

Exercise 33. ★

Prove that the following statements are true :

1. $\mathbb{Q}[\sqrt{3}] = \{a + b\sqrt{3} \mid a, b \in \mathbb{Q}\}$ is a subfield of $\mathbb{R}$.
2. $\mathbb{Q}[\sqrt{2}, \sqrt{5}, \sqrt{10}] = \{a + b\sqrt{2} + c\sqrt{5} + d\sqrt{10} \mid a, b, c, d \in \mathbb{Q}\}$ is a subfield of $\mathbb{R}$.
3. $\mathbb{Q}[i] = \{a + bi \mid a, b \in \mathbb{Q}\}$ is a subfield of $\mathbb{C}$.
4. $\{a + b\sqrt[3]{2} \mid a, b \in \mathbb{Q}\}$ is not a subfield of $\mathbb{R}$.
5. $\mathbb{Q}[i]$ and $\mathbb{Q}[j]$ are not isomorphic, where $j \in \mathbb{C}$ is a root of the polynomial $X^3 - 1$ different from 1, and $\mathbb{Q}[j] = \{a + b\sqrt{j} \mid a, b \in \mathbb{Q}\}$.

Citation tirée du livre "Géométrie" de Michèle Audin :

"Il faut faire des exercices. Il faut chercher les exercices. Un exercice n'est pas quelque chose dont il faut connaître <<la>> solution pour la réciter à un jury. Aucune notion ne peut être comprise ou assimilée sans un minimum de pratique, de recherche, d'échecs. **Un exercice sur lequel on n'a pas <<séché>> est un exercice inutile.**"